Autonomous Systems (AS)

"An AS is a group of IP networks operated by one or more network operator(s) that has a single and clearly defined external routing policy."

APNIC website

- Could be one or more companies/organizations but to the outside world looks like one collection of networks (routing prefixes).

- Each AS is assigned a 32 bit number for identification (abbreviated as ASN)
  - Older AS number was 16 bits
  - Assigned by: Internet Assigned Numbers Authority (IANA)

Autonomous Systems (AS)

- Why ASes
  -
  -
- ISP
  -
  -

Between ASs - Peering and Transit ASs

- Peering
  - ISP to ISP relationship
  - Two ISP provide cross connectivity to their transit customers
  - Peering point – the physical connection between two ISPs
  - IXP – Internet Exchange Point – physical infrastructure for peering connections

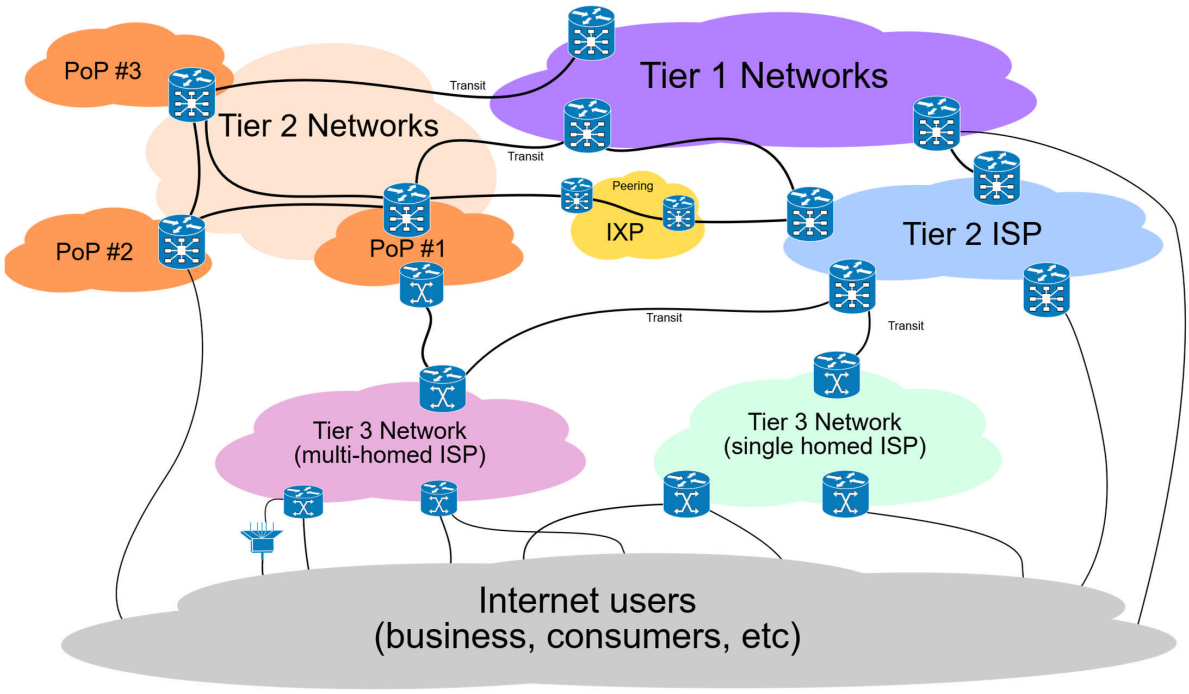
Types of ASes

- **Stub AS**
  - Single connection to outside world, does not carry anyone else's traffic
  - Only routes local traffic (either originated within the AS or destined for a network within the AS)

- **Multihomed AS**
  - Multiple connections to outside world
  - Used for redundancy
  - Will NOT carry traffic for others (is not a transit AS)

- **Transit AS** – carries transit traffic
  - An Internet Service Provider (ISP)
  - Will carry transit traffic (traffic that is just passing **through**)
  - ISP provides connectivity to the customer

AS Drawings



https://commons.wikimedia.org/wiki/File:Internet_Connectivity_Distribution_%26_Core.svg

Tier 1 and Tier 2 ISPs

- Tier 1 - Global reach
  - Peer with other Tier 1s typically do not pay
  - Backbone of the Internet
  - Have own infrastructure (undersea cables, cross country connections)

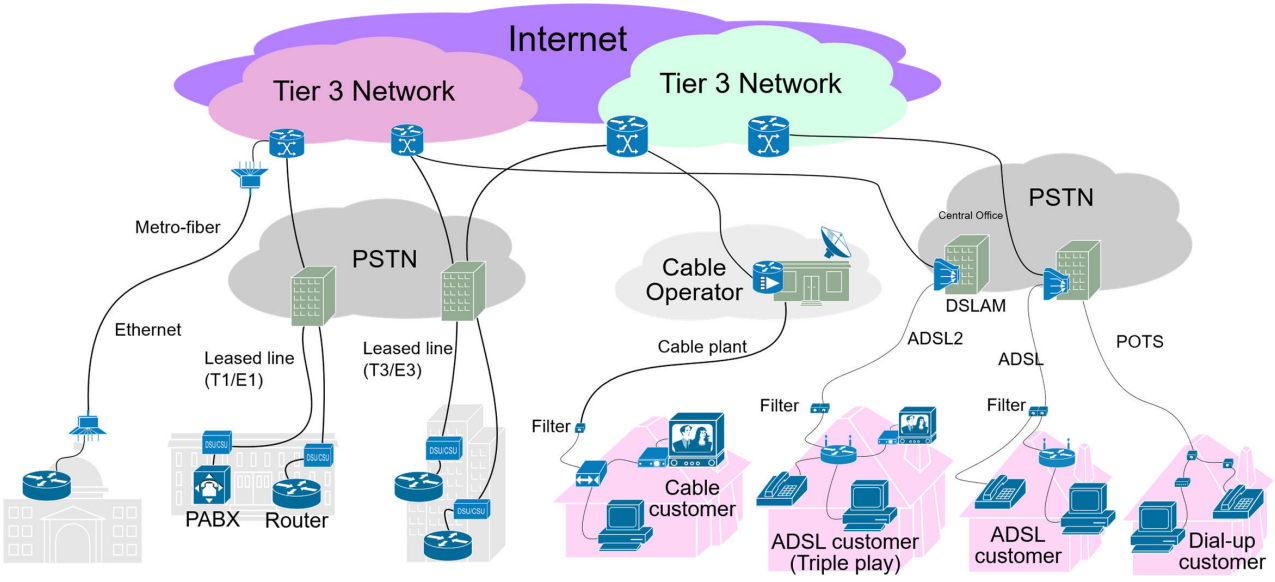
  - Examples:
    - Level 3 Communications
    - AT&T
    - Verizon
    - Lumen Technologies (CenturyLink)

Tier 1 and Tier 2 ISPs

- Tier 2 - Regional or National providers
  - They usually pay Tier 1 providers for connections
  - Peer with other Tier 2s with local or regional scope
  - Do not have the "global" reach of a Tier 1

  - Examples:
    - Comcast
    - Charter
    - Cox Communications
    - BT Group (UK)
    - Telstra (Australia)

Internet Service Providers



https://commons.wikimedia.org/wiki/File:Internet_Connectivity_Access_layer.svg

Cal Poly's AS

AS7960

California Polytechnic State University · calpoly.edu

AS7960 – California Polytechnic State University

Country	United States
Website	calpoly.edu
Hosted domains	30
Number of IPv4	72,448
Number of IPv6	0
ASN type	Education
Registry	ARIN
Allocated	5 years ago on Jan 17, 2019
Updated	5 years ago on Jan 17, 2019

IP Ranges — IPv4 Ranges | IPv6 Ranges

NETBLOCK	COMPANY	NUM OF IPS
129.65.0.0/16	California Polytechnic State University	65,536
198.188.150.0/23	California Polytechnic State University	512
198.188.152.0/22	California Polytechnic State University	1,024
198.188.159.0/24	California Polytechnic State University	256
207.62.152.0/24	California Polytechnic State University	256
207.62.153.0/24	California Polytechnic State University	256
207.62.154.0/24	California Polytechnic State University	256
207.62.155.0/24	California Polytechnic State University	256
207.62.156.0/22	California Polytechnic State University	1,024
207.62.160.0/24	California Polytechnic State University	256
207.62.161.0/24	California Polytechnic State University	256
207.62.162.0/24	California Polytechnic State University	256
[illegible]	[illegible]	[illegible]

Peers

There are 2 peers for this ASN.

PEERS	NAME
AS2152	California State University, Office of the Chancellor
AS3356	Level 3 Parent, LLC

https://ipinfo.io/AS7960

CSU Chancellor's office AS

AS number details

AS2152

California State University, Office of the Chancellor · cenic.org

CDN BITTORRENT

AS2152 – California State University, Office of the Chancellor

Country	United States
Website	cenic.org
Hosted domains	908
Number of IPv4	2,077,952
Number of IPv6	7.92×10^{28}
ASN type	Education
Registry	ARIN
Allocated	31 years ago on Jan 14, 1993
Updated	2 years ago on Jul 15, 2022

Peers

There are 169 peers for this ASN.

PEERS	NAME
AS101	University of Washington
AS10294	Coast Community College District
AS10310	Oath Holdings Inc

IP Ranges — IPv4 Ranges | IPv6 Ranges

NETBLOCK	COMPANY	NUM OF IPS
104.249.91.0/24	Orange County Department of Education	256
104.249.94.0/24	Orange County Department of Education	256
130.150.0.0/16	California State University, Office of the Chancellor	65,536
130.166.51.0/24	California State University, Northridge	256
130.17.0.0/16	California State University, Stanislaus	65,536
130.182.0.0/16	California State University, Los Angeles	65,536
130.191.0.0/16	San Diego State University	65,536

Downstreams

There are 87 downstreams for this ASN.

DOWNSTREAMS	NAME
AS10294	Coast Community College District
AS10867	Foothill-DeAnza Community College District
AS11159	Los Angeles Public Library
AS7960	California Polytechnic State University

https://ipinfo.io/AS2152

Alabama Supercomputer AS

AS number details

AS3464

Alabama Supercomputer Authority · asc.edu

CDN BITTORRENT

AS3464 – Alabama Supercomputer Authority

Country	United States
Website	asc.edu
Hosted domains	199
Number of IPv4	210,432
Number of IPv6	7.92×10^{28}
ASN type	Education
Registry	ARIN
Allocated	30 years ago on Mar 04, 1994
Updated	2 months ago on Nov 30, 2023

IP Ranges IPv4 Ranges IPv6 Ran

NETBLOCK	COMPANY	NUM OF IPS
129.66.0.0/16	Alabama Supercomputer Authority	65,536
157.149.0.0/16	Alabama State Department of Education	65,536
192.111.112.0/24	Alabama State University	256
192.211.32.0/23	Alabama Industrial Development Training	512
192.211.39.0/24	Alabama Industrial Development Training	256
192.211.41.0/24	Alabama Industrial Development Training	256
192.211.42.0/24	Alabama Industrial Development Training	256

Peers

There are 7 peers for this ASN.

PEERS	NAME
AS1031	Peer 1 Internet Service LLC
AS10490	Georgia Institute of Technology
AS174	Cogent Communications
AS18682	Alabama A&M University
AS40204	Jacksonville State University
AS6939	Hurricane Electric LLC

https://ipinfo.io/AS3464

Different Types of Routing Protocols (IGP, EGP)

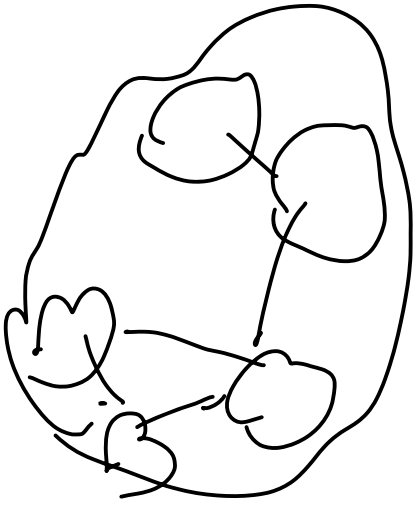
Interior Gateway Protocol w/in a cloud (within AS)

Exterior Gateway Protocol outside of AS

Dynamic Routing Protocols { OSPF, RIP → in lab

IP is active,

Ethernet is passive



IP Routing Protocols – Within and Between ASes

- Multiple standard routing protocols
  - **Interior routing protocols**
    - May be called Interior Gateway Protocols (IGP)
    - OSPF (interior, may be used in an ISP AS or company network)
    - IS-IS (typically used in ISP ASes) → used by large organizations
    - RIPv2 (typically used within a company network)
  - **Exterior routing protocols**
    - Border Gateway Protocol (BGP)
      - EBGP –Exterior Border Gateway Protocol
  - **Route Redistribution**
    - Used to redistribute BGP routes within the AS
    - iBGP – modified BGP is one example

General Routing Issues To Consider

- Routing – forward out on the correct interface towards the destination
  - Use (creates) : the routing table
  - Performance Criteria (cost)

    Hop-count
  - When and how to make routing decisions:
    - When: when packet arrives
    - Where: in the router (based on routing table)

Shortest Path Algorithms

- **Shortest Path Routing Goal**: Find the least cost route from the source to the destination
  - In effect creates a logical shortest path tree topology over the physical topology (minimum cost spanning tree)
- Using some measurement of "cost" (also call metric)
  - Hop count
  - Link Bandwidth
  - Delay
  - Current utilization
- Two Algorithms
  - Dijkstra's
  - Bellman-Ford

Process to Develop Routing Table

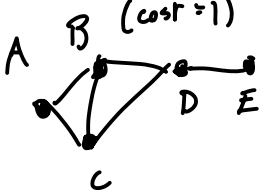
- Collect Network Info → Process → Routing Table
  (create a data structure)
- Minimum spanning tree with the router as the root

Graphs – A quick review

- Adjacency Matrix

	A	B	C	D	E
A		1	1		
B	1		1	1	
C	1	1		1	
D		1			1
E				1	

simple graph (cost = 1)



- Weighted Graphs and Directed Weighted Graphs

	A	B	C	D	E
A		4	3		
B	4		1	3	
C	3	1		1	
D		2	1		2
E				2	

Distance Vector Routing

- Known as: **Distance Vector Routing**

- Generates a next hop count table

Dest Subnet	Hop Count	Next Hop

- Who to share with:

- What to share:

Distant Vector Routing (e.g. RIP) Con't

- Uses "Bellman-Ford's" shortest path algorithm
- Each router only has a local view of the network
- Example implementations
  - RIP – Router Information Protocol
  - RIP2
- Issues
  - Scalability
  - Count to Infinity problem (loops)

Link State Routing

- Known as: **Link State Routing**
- Generate a **link state database,** which is a weighted adjacency matrix but it is called a **<u>link state database</u>**)
- Who to share with:
- What to share:

Link State Routing (e.g. OSPF) Con't

- Uses Dijkstra's Shortest Path Algorithm
- Each router has a global view of the network
- Example implementations
  - OSPF
- Issues
  - Complexity

Routing Information Protocol v2 (RIPv2)

- RIPv2
  - Interior gateway protocol (so within an Autonomous System)
  - Distant Vector routing protocol
  - Supports subnetting and CIDR addressing
    - Including Variable Length subnet masking (VLSM)
  - Cost Metric
    - Hop count only
    - 15 is maximum hop count (so 16 = infinity)
  - Updates
    - Sent every 30 seconds
    - Also triggered by changes

Open Shortest Path First (OSPF)

- **Open Shortest Path First** (RFC 2328)
  - Interior gateway protocol (so within an Autonomous System)
  - Shortest path calculation:
    - Link State
    - Dijkstra's shortest path algorithm to calculate routing table
  - Features
    - Multiple distance metrics
    - Dynamic
    - QoS
    - Load Balancing
    - Hierarchical (supports Areas – next slide)
    - CIDR support

OSPF and Areas

OSPF (cont.)- Areas

- OSPF Hierarchical routing
  - Supports an abstraction called Areas
    - Allows for a AS to be further subdivided into areas
    - Routers within an Area exchange routing information with each other
    - Routers external to Area only share routing information with Area edge routers
    - Allows for hierarchical routing within an AS

Interior and Exterior Routing Protocols

- **Interior Gateway Protocol** (IGP)
  - Also called: Interior Routing
  - Sharing routes **within** a AS or Company Network
  - Routing Protocols: OSPF, IS-IS RIPv2, EIGRP

- **Exterior Gateway Protocol** (EGP)
  - Allow two ASs to exchange routing information.
  - Sharing routes **between** AS
  - Main protocol: BGP
  - For external routes its EBGP (Exterior BGP)
  - Must have an AS Number to use EBGP

Exterior Gateway Routing Protocol

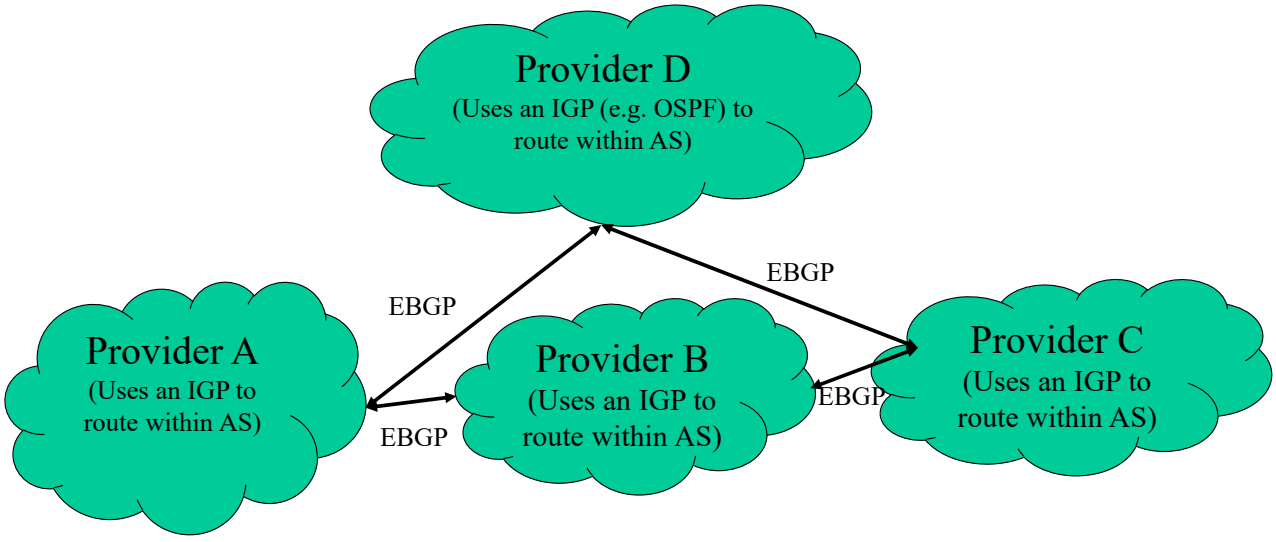
- Border Gateway Protocol (BGP) see RFC 1771
  - EBGP – Exterior Border Gateway Protocol
    - Speaks BGP between ASes.
  - IBGP – Interior BGP
    - Speaks BGP with other BGP speakers in the AS

- Used to route between ASes
  - Exchanges network reachability information
    - Says you can get to network X via this path of ASes

  - Needs to be concerned with policies (business and political)
  - Uses a modified Distance Vector Protocol – Path-Vector Routing

  - Runs over a reliable transport protocol (TCP)

Two Different Types of Routing Protocols (IGP vs. EGP)



Key Points:
- Each provider uses an **IGP** to learn routes **within** the AS
- Each provider uses **BPG** to learn routes **between** ASes

EBGP (cont.)

- **BGP is about route generation <u>between</u> ASes**
  - EBGP – Talk to external BGP routers
  - IBGP – Talk to internal BGP routers

- **Only exchanges routing information at the AS level**
  - Not routing info about the internals of the ASes
  - Exchanges full **AS path info** to destination IP

    - 5.0.0.0/8: <AS 4, AS 9, AS 7, AS 3>

BGP Route Sharing and BGP Speakers

- **Exchanges the entire route from source to destination**
  - Routing decision can be made based on the entire route

  - **Why entire route**
    - Allows for policy decisions to be made on the possible routes to a network
    - Prevents loops in the routing path

- Note on BGP Speakers
  - Any router that speaks BGP to other routers is a BGP Speaker
  - So an ASBR is a BGP speaker
  - A BGP speaker may or may not speak outside the AS

BGP Route Generation

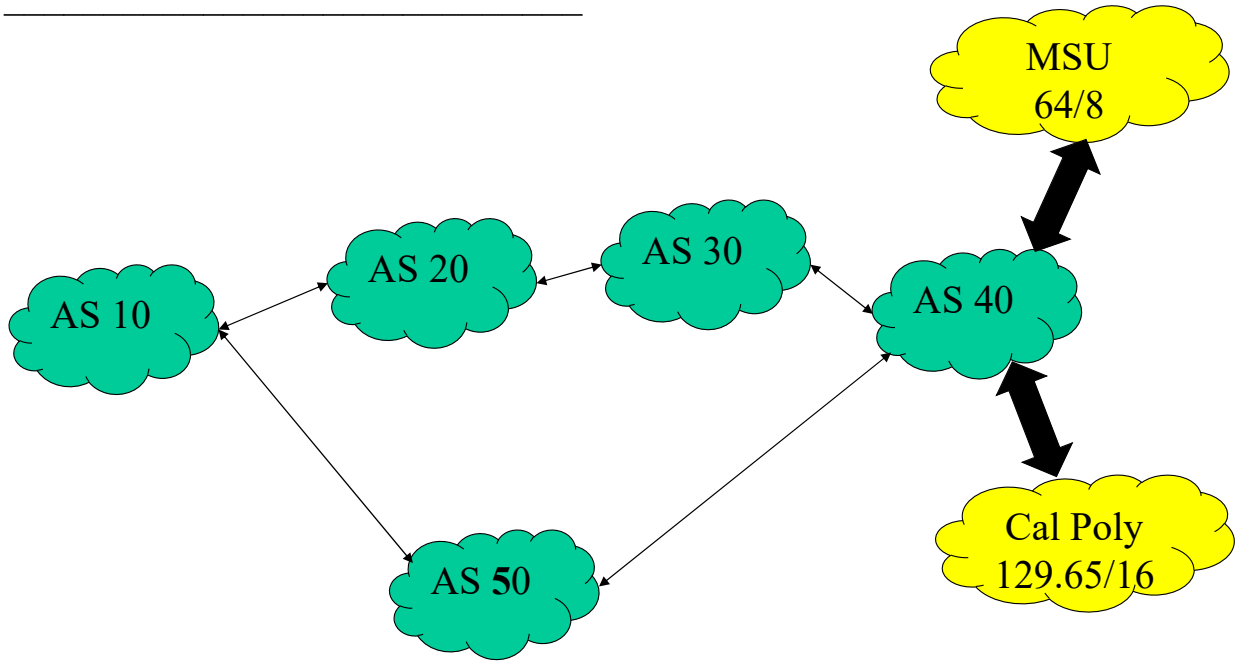
1. Receive reachability information from connected BGP Speakers
   - Receives from other ASes' BGP-speaking edge router
   - The Route is the entire AS path for each destination network

2. Apply Policy to the routes and rank the routes

3. Pick the route to each network based on the ranking of the routes and the shortest path based on AS count
   - The distant vector route calculation is based on the number of ASes transited
   - Store the routes in the local Router Information Base (RIB)

4. Sharing information (route redistribution)
   - Share route information to internal BGP Speakers (using IBGP)
   - Share route information with IGP routers (e.g. via OSPF)

BGP Example



BGP Example – Space for notes

Routing Summary

- Interior Routing protocols within an AS (e.g. RIP, OSPF)
  - OSPF Areas are subsets of an AS
- BGP
  - Shares reachability information between ASes
  - iBGP – Route redistribution to other **internal** BPG speakers
  - Allows routers in an AS to receive information about exterior networks
- Route redistribution
  - Share external routes with internal routers

Routing Table Creation

- Run routing protocols which are input to the RIB

- RIB – Router Information Base (control plane)
  - Routes learned by routing protocols
    - OSPF, RIP, route redistribution, iBGP, BGP
  - Also includes Static routes and directly connected routes
  - Cisco Command: **show ip route**

- FIB – Forward Information Base (data plane)
  - Optimized selection of routes based on RIB
  - Searched using hardware on higher end routers
  - Includes the actual route that will be taken (so if redundant routes, the one being used will be listed)
  - Cisco Command: **show ip cef** (Cisco Express Forwarding)

16-bit and 32-bit assigned AS Numbers

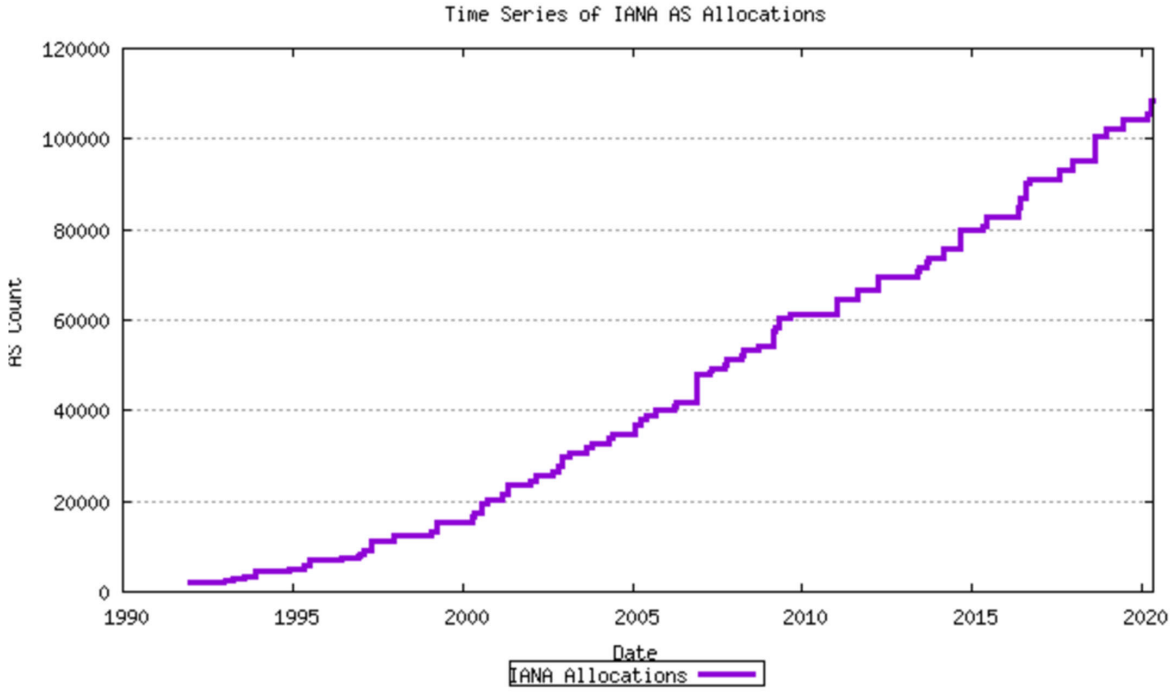
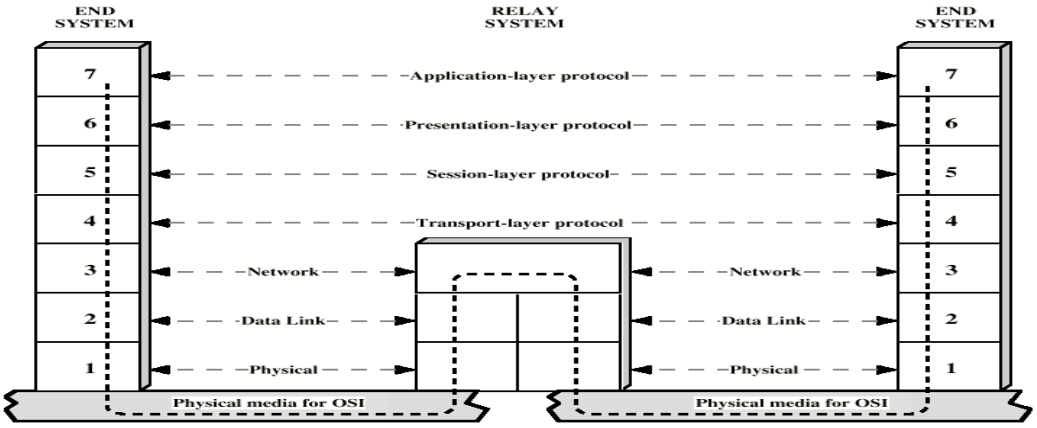


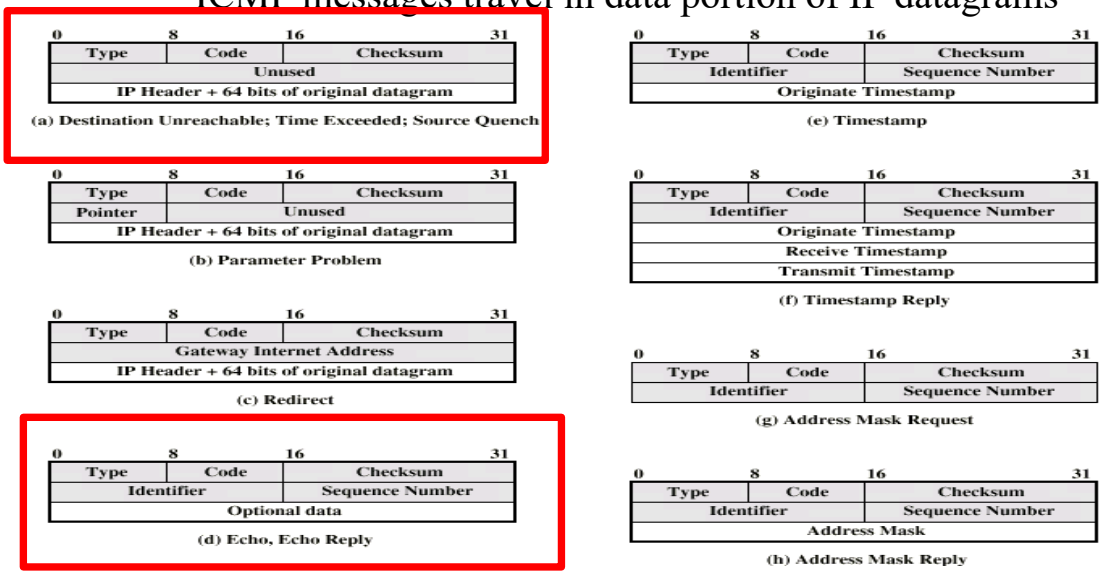
Figure 2 - Cumulative IANA AS block allocations

https://www.potaroo.net/tools/asn32/

Routers – why they are called layer 3 devices

- Routers - operate on the first 3 layers
  - Bits are received on the physical medium
  - The Frame is passed up to layer 2 (e.g. Ethernet) – layer 2 header removed
  - Datagram is passed to IP layer (layer 3)
    - Layer 3 makes routing decision (picks outgoing interface)



Internet Control Message Protocol

- ICMP
  - RFC 792
  - Provides communication between IP on one machine and IP on another machine
  - Used to send error indication messages to the source IP
  - ICMP messages travel in data portion of IP datagrams